Paper References

Sr. No	Title of Paper	Author	Remark	Future Scope	Link
1	**Automatic Solar Tracking System**	• Mohan S • Rajkumar. K • Rajakumar. P • Hari Pradosh S M • Gandhi S • Balasakthishwaran M	The solar tracking components used and source angles are mentioned for the source outcomes of the servo-motor rotation.	A better tracking system can be implemented with the help of the models used for the calculative assumptions and the outcome models expressed with the help of multiple softwares programs.	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10047670
2	**A Comparative Evaluation of Various Solar Trackers to Harness Maximum Energy**	• S. Venkateshwarlu • V. Sree Pranav • Ch. Sai Anirudh • K. Shashidhar Reddy	Various types of trackers, their characteristics, and comparisons between them are provided.	The comparative analysis taking place in this paper prints out a clear image of the pro's and cons of the various solar trackers In addition, the data in this paper can be	https://ieeexplore.ieee.orPanelsg/stamp/stamp.jsp?tp=&arnumber=10125472

				used for further improvement of the entire system of tracking procedures	
3	**Single-Axis Solar Tracker for Solar Panels with Power and Cost Analysis**	• Daniel Kris Hamonangan Sitompul • Oyas Wahyunggoro • Adha Imam Cahyadi	Power analysis by using mathematical modules as well as cost-effective analysis of making the initial investment back and pros of using single-axis solar tracker	Can be used to implement cost-effective solar power production operations with complete financial as well as theoretical clarity over the said product	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9807779
4	**Design of Standalone Solar PV System Using MPPT Controller and Self-Cleaning Dual Axis Tracker**	• Yash Ajgaonkar • Mayuri Bhirud • Poornima Rao	Integration of MPPT controller and P&O technique Inbuilt self-cleaning feature.	Incorporate Internet of Things (IoT) technology to enable real-time data collection and analysis. This can provide insights into system performance, enabling	https://ieeexplore.ieee.org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=8728494

				proactive maintenance and optimization.	
5	**The Effective Solar Panel Tracker to Obtain Maximum Energy Optimization by Using Innovative Machine Learning**	• Rajanish Kumar Kaushal • V. Velmurugan • S. Mahendran • Arul Prakash M • Balvender Singh • Joshuva Arockia Dhanraj	Equation and optimization of energy through use of exciting and new machine learning technique		https://ieeexplore.ieee.org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=10084249
6	**Simulation Model of Autonomous Solar Power Plant with Dual-Axis Solar Tracker**	• S.V. Mitrofanov • D.K. Baykasenov • M.A. Suleev	Understanding of the entire solar tracking system and its input and output characteristics for an entire power plant made possible through use of software (MATLAB/Simulink)	Better outcomes are to be derived from the simulated results achieved in this paper. The data is crucial for understanding the system and process for better real-life implementation.	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=8544275

7	**Design of a GPS-Based Solar Tracker System for a Vertical Solar Still**	• Dian Artanto • A. Prasetyadi • Doddy Purwadianta • Rusdi Sambada	Methods of using the solar tracking system for better efficiency and further optimization of the harnessed energy for vertical solar stills (desalination of water plant) as well as data collection	The usage of gps technique for better collection of solar energy through solar rays (sunlight) the generated energy is used to harness a vertical solar still (desalination water plant) the outcomes can be used for further projects through the required understanding of the inputs and outputs of the required energy for the outcome system	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=7885780

8	**Algorithm for Single Axis Solar Tracker**	• Nedyalko Todorov Katrandzhiev • Nikolay Nenkov Karnobatev	The Algorithm is explained on the basis of the formula of math (Meinel model, Energy increase calculation) for the better understandings of an outcome a reference can be made	The algorithm can be better implemented with better understanding of the softwares used while models used for calculation are critical for the solar tracking process makng it more efficient	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=8549644
9	**Review of Solar Tracker Technologies**	• Sami Attia • Ryadh Al-MahruqiDiscuss the environmental impact of solar tracking systems, including considerations related to land use, wildlife, and ecosystem sustainability. Highlight research on eco-friendly tracking	This review paper provides a comprehensive overview of solar tracker technologies, discussing various tracking systems, their advantages, and the state of the art in solar tracking research.	· Discuss the environmental impact of solar tracking systems, including considerations related to land use, wildlife, and ecosystem sustainability. Highlight research on eco-friendly tracking	https://www.researchgate.net/publication/340084108_Technologies_of_solar_tracking_systems_A_review

10	**Design and Implementation of a Solar Tracking System with Image-based Data Acquisition**	• Robert W. Kost • Seyed B. M. Iman • Robert S. Balog	This paper discusses the design and implementation of a solar tracking system that utilizes image-based data acquisition for precise solar panel alignment.	Consider using cloud-based computing for image analysis and tracking calculations. This can provide scalability and reduce the processing load on local hardware.	(PDF) Design and Implementation of a Solar-Tracking Algorithm
11	**Role of geoinformatics in ensuring global food security with special reference to dryland agroecosystems**	• Chandrashekhar M. Biradar • Xiangming Xiao • William Payne • Prasanna H. Gowda	Agriculture is crucial for food security in developing countries, and better agricultural information is essential for understanding trends. Geospatial technologies, like remote sensing	Continuously update and adopt advanced remote sensing technologies, such as high-resolution satellite imagery and unmanned aerial vehicles (UAVs), to	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=6621918

			and GPS, are transforming food security research, policy, and outreach. The International Center for Agricultural Research in Dry Areas is leading this effort.	monitor crop health, land use, and environmental factors more effectively.	
12	**A multiple representation data structure of 3D building textures**	• Bo Mao • Yifang Ban	TextureTree simplifies 3D city model visualization by storing building textures in different LoDs, segmenting them by color difference, and creating a multiple representation data structure, reducing data volume while maintaining visual	Ensure that TextureTree can be easily integrated with various 3D modeling software, GIS platforms, and geospatial data standards for seamless data exchange and compatibility.	https://ieeexplore-ieee-org.library.somaiya.edu/document/5981035

			similarity.		
13	**Arduino based Dual Axis Solar Tracker**	• Akash Singh • Swapnil Adhav • Anirudha Dalvi • Atharva Chippa • Milind Rane	This research paper discusses the development and design of a dual-axis solar panel tracker, aimed at improving performance ratios and maximizing solar energy collection from photovoltaic panels. It aims to offer more efficient methods for solar energy generation.	Implement systems for early fault detection and predictive maintenance. This can help prevent system breakdowns and optimize long-term performance.	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=9742876
14	**Geoinformatics: A Tool for Environmental Management**	• S.S. Negi • M.S. Meena	This paper discusses the role of geoinformatics in environmental management and highlights its applications in		https://www.arcjournals.org/pdfs/ijres/v5-i1/2.pdf

			monitoring and decision-making.		
15	**A Minutes-Based Metric System For Geographic Coordinates in Mobile GIS**	• Mohamed Eleiche	This paper analyzes the complexity of the ancient Babylonian sexagesimal system for geographic position, proposing a new modern metric system based on minutes for mobile GIS. The system, which uses degrees, minutes, and seconds, is not suitable for mobile users' needs.	Ensure that the new metric system is compatible with and leverages modern technologies, including GPS, mapping services, and geospatial data collection tools commonly used in mobile GIS.	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=5567792

16	**Augmenting Policy Making for Autonomous Vehicles through Geoinformatics and Psychographics**	• Feras A. Batarseh • Shruthi Chanthati • Nithin Kotte • Abhinav Kumar	This manuscript uses descriptive and predictive analytics to evaluate and challenge common beliefs about Autonomous Vehicles (AVs) safety and collision causes, aiming to inform policy making for these new technologies.	Conduct a long-term study to track the evolution of AV safety over time. This will help identify trends and assess the effectiveness of policy changes.	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=8475980
17	**Practical Implementation of Dual Axis Solar Power Tracking System**	• Hend Abd El-monem Salama • Adel Taha Mohamed Taha	This lab-developed solar system uses a dual-axis tracking module, connected to an Arduino program, and a maximum power point tracking (MPPT) solar	Implement a data monitoring system that records solar panel performance, environmental conditions, and energy generation. Analyze this data	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=8635197

			model for the lowest cost.	to identify areas for improvement and better understand system behavior.	
18	**Dual Axis Solar Tracker With Fuzzy Logic Method**	• Yustinus Bagus Arisotya • A. Bayu Primawan • Djoko Untoro Suwarno	The research aims to improve solar trackers using LDR and sun movement calculations by creating a two-axis solar tracker using fuzzy logic. The tracker uses two stepper motors and communicates with the Arduino Mega and ESP-12E, enabling it to adapt to bad weather conditions.	Develop a reliable communication system between the solar tracker and the ESP-12E module. Test the communication under various scenarios to ensure data exchange reliability.	https://ieeexplore-ieee-org.library.somaiya.edu/stamp/stamp.jsp?tp=&arnumber=10010188